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OpenSim Creator: A Graphical User Interface for Building OpenSim Musculoskeletal Models

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Summary

OpenSim Creator is an open-source "biomechanical CAD" user interface (UI) that focuses on building and evaluating musculoskeletal models in OpenSim, something that previously required manually editing extensible markup language (XML). In this paper, we outline its architecture and provide concrete examples of how OpenSim Creator has already been used to build new biomechanical models.

Statement of need

OpenSim (Delp et al., 2007; Seth et al., 2018) is an open-source musculoskeletal modeling platform with a large and growing number of users in diverse fields. This follows the general trend of musculoskeletal modeling being increasingly used in clinical (Marconi et al., 2023), biomechanical (Porsa et al., 2016), and palaeobiological (Beesel et al., 2021; Demuth et al., 2023) studies. However, creating a biomechanical model remains a challenge because the process is both laborious, as it requires the input of a large amount of biological data, and complex, as it requires expertise in biomedical imaging, model/data file formats, and musculoskeletal anatomy.

As an illustrative example, Bishop et al. (2021) outline a workflow for building a biomechanical model of a dinosaur. It involves specialized steps such as a) acquiring and segmenting bone morphology, centers of mass, and inertia tensors from fossil records; b) rigging the bones into a skeletal model; c) reconstructing musculature from soft tissue volumes; and d) using the resulting model in a variety of biomechanical simulations. Specialized procedures like these may need to be repeated for each subject in an experiment because animals are unique, which can significantly increase the amount of labor required. Scaling existing models can minimize that work, but designing a correct, unbiased (van der Kruk, 2025), scaling procedure that is capable of handling a variety of shapes, injuries, and deformities is also challenging.

Similar challenges have been addressed in fields such as 3D design, engineering, and game development. For example, engineers commonly use computer-aided design (CAD) software both to reduce labor, by automating actions such as 3D object placement; and to increase the user's expertise, by enabling model editing in a safe undoable area with immediate, useful, visual feedback. As musculoskeletal modelling with OpenSim expands to fields that are not traditionally occupied by engineers (e.g., medicine, palaeobiology), real-time model development with visual feedback becomes paramount. OpenSim Creator provides that tooling by building on ideas from existing CAD software - with a biomechanical twist.

Design and implementation

We initially studied existing open-source tools (e.g., Blender (2018)) to find common design and architectural themes. We found that, in most cases, they focus on providing immediate visual feedback whenever a user performs an action and are continuously improved over many releases. Immediate feedback is usually achieved by directly integrating the UI with the underlying state/model representation (as opposed to relying on an indirect mechanism, such as a remote procedure call). Based on that, and our previous experience with OpenSim, we decided that OpenSim Creator's architecture should focus on the following four key attributes:

- **High performance**: to enable real-time model editing and instant feedback.
- **Low-level UI control**: to enable a highly customized user experience. Specialized tasks in biomechanical model building (e.g. landmark placement, mesh reorientation, mesh warping, real-time data visualization) can require low-level control over the rendering process.
- **Direct integration with the OpenSim API**: to directly access and visualize biomechanical model quantities (e.g. muscle lengths, moment arms) on-demand with minimal development or performance overhead.
- **Iterative software, and model, development**: to enable rapid software prototyping and feedback from the research community.

To facilitate those requirements, OpenSim Creator took inspiration from the games industry, which is commonly associated with rapid prototyping and real-time rendering. We developed OpenSim Creator using a combination of C++, a high-performance language that directly integrates with the OpenSim API; a custom OpenGL renderer, to tightly control real-time 3D editing; and Dear ImGui (Cornut (2014)), which provides customizable low-level 2D UI.

Organizationally, we adopted an open, iterative, development methodology. All releases of OpenSim Creator have been available on GitHub with an Apache 2.0 license as early as the first month of development, with new releases published every 1-2 months. As a consequence, we have been able to continuously receive—and implement—feedback from researchers in the community over 36 release cycles. This is in addition to a healthy supply of real model-building challenges from our own research, which contributed heavily to OpenSim Creator's design and architecture (e.g. Figure 1).

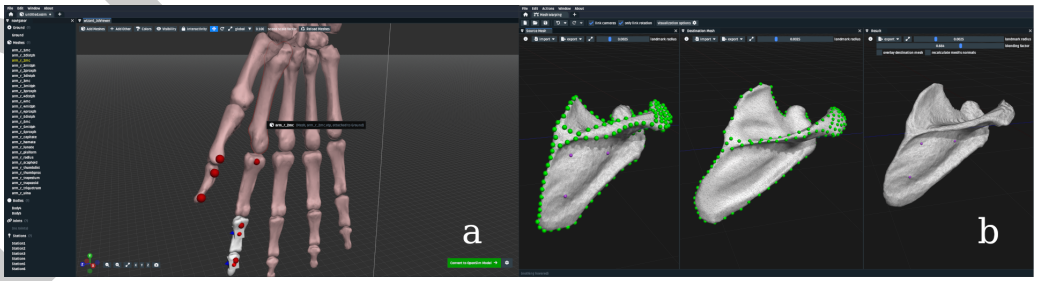

Figure 1: Screenshots of OpenSim Creator's specialized workflows. a) The mesh importer workflow, which focuses on reorienting and rigging meshes in a free-form editor, b) the mesh warper workflow, which focuses on placing corresponding landmarks on meshes (left and middle) with real-time updates to a Thin-Plate Spline (Bookstein, 1989) warped mesh (right).

Example use-cases

Subject-specific models are necessary because all individuals are unique and can have a wide variety of injuries and deformities. Our goal was to create a subject-specific model of a human shoulder. A previously published (Seth et al., 2019) generic shoulder model was used as a template by superimposing it over subject-specific CT scans in OpenSim Creator's mesh

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importer UI (Figure 1a). The mesh importer UI enabled real-time, free-form placement and orientation of the subject-specific joints and bodies. The scene was automatically converted into an OpenSim model. Muscle volumes from subject-specific MRI data were combined with moment arm data from the generic model to create a line of action for each muscle in the model. The model editor UI provided real-time visual feedback about how each muscle's moment arm was affected by the placement of attachment points and wrapping surfaces on the line of action within the model (Figure 2).

Another study evaluated how a given biomechanical model simulated a sit-to-stand movement. The predictive simulations proved to be computationally expensive, and OpenSim Creator was used to simplify the existing model by eliminating wrapping surfaces and leveraging quick visual feedback (both as 3D renders and 2D plots) to place muscle via points while preserving the moment-arms of the original model (Heijerman (2023)).

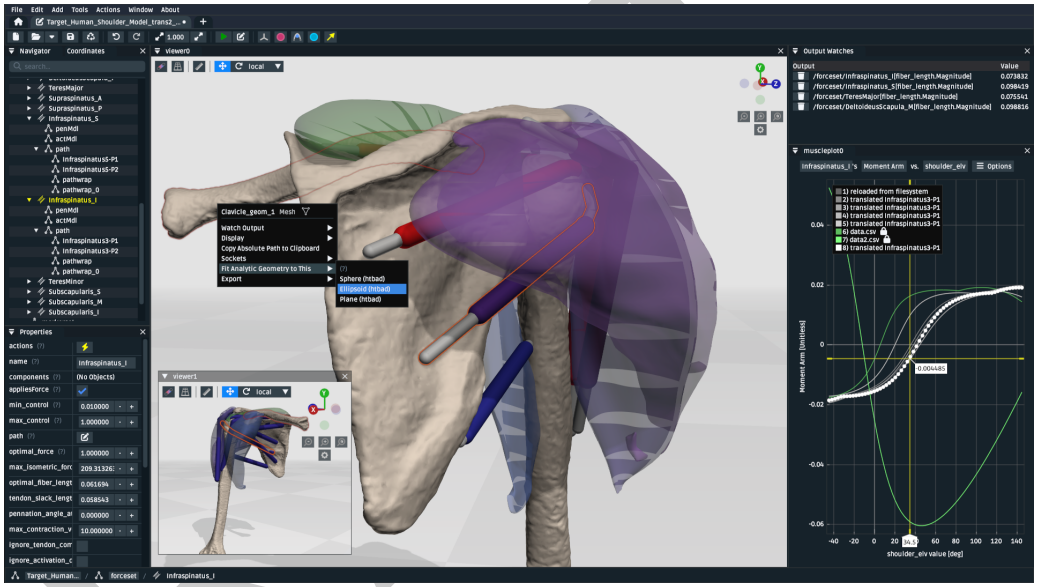

Figure 2: A screenshot of OpenSim Creator's OpenSim model editor UI, showing a subject-specific human shoulder model made using it. The model editor can host multiple panels, each of which are updated whenever the user edits the model. For example, the muscle plotter (right) can be used to show how a muscle's moment arm changes as a user edits the muscle in the 3D viewport (middle) or in the properties panel (left).

Conclusions

By combining a rapid-prototyping architecture inspired by the games industry with a development approach that continuously incorporates feedback from the biomechanical research community, we designed and developed a graphical UI that enables the creation and evaluation of OpenSim models more quickly and easily.

OpenSim Creator's design and architecture enables it to render multiple high-vertex-count bone models in real-time on a mid-range laptop. Its control over the UI enables customized workflows where researchers can create models visually, and its tight integration with OpenSim enables it to show moment arms in real time as model creation proceeds. Our users report significantly reduced amounts of time and effort to build, edit, and evaluate models when compared to existing approaches.

As software constantly evolves, we are looking forward and anticipate leveraging OpenSim Creator to create specialized workflows for visually-assisted model warping, real-time sensor visualization, and the import of other biomedical data, in addition to continuing to improve the

day-to-day user experience of creating an OpenSim model from medical imaging and emerging sources of anatomical structure and material properties.

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References

Beesel, J. van, Hutchinson, J. R., Hublin, J.-J., & Melillo, S. M. (2021). Exploring the functional morphology of the gorilla shoulder through musculoskeletal modelling. *Journal of Anatomy*, *239*(1), 207–227. https://doi.org/10.1111/joa.13412

Bishop, P. J., Cuff, A. R., & Hutchinson, J. R. (2021). How to build a dinosaur: Musculoskeletal modeling and simulation of locomotor biomechanics in extinct animals. *Paleobiology*, *47*(1), 1–38. https://doi.org/10.1017/pab.2020.46

Blender, O. C. (2018). *Blender - a 3D modelling and rendering package*. Blender Foundation. http://www.blender.org

Bookstein, F. L. (1989). Principal warps: Thin-plate splines and the decomposition of deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *11*(6), 567–585. https://doi.org/10.1109/34.24792

Cornut, O. (2014). *Dear ImGui: Bloat-free graphical user interface for c++ with minimal dependencies*. https://github.com/ocornut/imgui

Delp, S. L., Anderson, F. C., Arnold, A. S., Loan, P., Habib, A., John, C. T., Guendelman, E., & Thelen, D. G. (2007). OpenSim: Open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, *54*(11), 1940–1950. https://doi.org/10.1109/TBME.2007.901024

Demuth, O. E., Wiseman, A. L. A., & Hutchinson, J. R. (2023). Quantitative biomechanical assessment of locomotor capabilities of the stem archosaur euparkeria capensis. *Royal Society Open Science*, *10*(1), 221195. https://doi.org/10.1098/rsos.221195

Heijerman, S. (2023). *Addition of torso musculature to a simplified musculoskeletal model for 2D predictive simulation of the sit-to-walk movement*. http://resolver.tudelft.nl/uuid:91925ea8-199e-4304-8b8f-8f52519a7931

Marconi, G., Gopalai, A. A., & Chauhan, S. (2023). Effects of powered ankle-foot orthoses mass distribution on lower limb muscle forces–a simulation study. *Medical & Biological Engineering & Computing*, *61*(5), 1167–1182. https://doi.org/10.1007/s11517-023-02778-2

Porsa, S., Lin, Y.-C., & Pandy, M. G. (2016). Direct methods for predicting movement biomechanics based upon optimal control theory with implementation in OpenSim. *Annals of Biomedical Engineering*, *44*(8), 2542–2557. https://doi.org/10.1007/s10439-015-1538-6

Seth, A., Dong, M., Matias, R., & Delp, S. (2019). Muscle contributions to upper-extremity movement and work from a musculoskeletal model of the human shoulder. *Frontiers in Neurorobotics*, *13*. https://doi.org/10.3389/fnbot.2019.00090

Seth, A., Hicks, J. L., Uchida, T. K., Habib, A., Dembia, C. L., Dunne, J. J., Ong, C. F.,

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DeMers, M. S., Rajagopal, A., Millard, M., Hamner, S. R., Arnold, E. M., Yong, J. R., Lakshmikanth, S. K., Sherman, M. A., Ku, J. P., & Delp, S. L. (2018). OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement. *PLOS Computational Biology*, *14*(7), 1–20. https://doi.org/10.1371/journal.pcbi.1006223

van der Kruk, E. (2025). BIASMECHANICS: Does an unconscious bias still persist in biomechanics, positioning males as the default in human research? A meta-analysis on the journal of biomechanics 2024 publications. *Journal of Biomechanics*, *181*, 112560. https://doi.org/10.1016/j.jbiomech.2025.112560

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